

A Literature-Implied Answer to the Schur/RNP Question for Lipschitz-Free Spaces

Math discovery packet

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1 Status

This is a lightweight *literature-implied answer*, not an original solution. The source paper is M. Cuth, M. Doucha and P. Wojtaszczyk, *Embeddings of Lipschitz-free spaces into ℓ_1* , arXiv:1909.05285. The supporting paper is R. J. Aliaga, C. Gartland, C. Petitjean and A. Proc-hazka, *Purely 1-unrectifiable metric spaces and locally flat Lipschitz functions*, arXiv:2103.09370.

2 Source Question

The source paper first asks, in Question 1 of the introduction, whether every closed negligible subset M of an $\mathbb{R}$ -tree has $\mathcal{F}(M)$ isometrically embeddable into some $\ell_1(\Gamma)$. That question is answered negatively inside the same paper by Theorem 1, so it is not recorded as a separate literature answer.

The surviving external question appears after Corollary 2.4 (`c:RNPSchurL1`). There the source paper proves that, for a closed subset M of an $\mathbb{R}$ -tree, the following are equivalent:

1. M is negligible,
2. $\mathcal{F}(M)$ has the Schur property,
3. $\mathcal{F}(M)$ has the Radon–Nikodym property,
4. $\mathcal{F}(M)$ contains no isomorphic copy of L_1 .

It then states that it is unknown whether any of the corresponding equivalences, in particular Schur $\iff$ Radon–Nikodym, hold for general Lipschitz-free spaces.

3 Supporting Theorem

ArXiv:2103.09370 proves the broader theorem. Theorem C in its introduction, and Theorem 4.6 (`thm:equivalences`) in the body, state that for every metric space M the following conditions are equivalent:

1. the completion of M is purely 1-unrectifiable,
2. $\mathcal{F}(M)$ has the Radon–Nikodym property,
3. $\mathcal{F}(M)$ has the Krein–Milman property,

4. $\mathcal{F}(M)$ has the Schur property,
5. $\mathcal{F}(M)$ contains no isomorphic copy of L_1 .

The body theorem also includes the equivalent quantitative formulation that $\mathcal{F}(M)$ contains no $(1 + \varepsilon)$ -isomorphic copy of L_1 for some $\varepsilon > 0$.

4 Identification

The free space over a metric space and over its completion are canonically the same for the purposes of the theorem, as noted at the start of the proof of Theorem 4.6 in the supporting paper. Therefore the implications

$$\mathcal{F}(M) \text{ has the Schur property} \iff \mathcal{F}(M) \text{ has the Radon–Nikodym property}$$

hold for every Lipschitz-free space. This gives an affirmative answer to the general Schur/RNP equivalence question left open after Corollary 2.4 of arXiv:1909.05285.

5 Provenance and Scope

This packet is placed in `literature_implied_answers`. The supporting paper explicitly treats the broad Schur/RNP problem for Lipschitz-free spaces and says the equivalence was suggested by previous work, but a source search did not find an explicit citation to arXiv:1909.05285 or to the exact sentence in that paper. Hence the relation is recorded as an agent-identified consequence of a decisive later theorem.

No part of this packet claims a new proof. Its role is duplicate memory: future agents should not attack the Schur/RNP equivalence for all Lipschitz-free spaces as open.

6 References

- M. Cuth, M. Doucha and P. Wojtaszczyk, *Embeddings of Lipschitz-free spaces into ℓ_1* , arXiv:1909.05285.
- R. J. Aliaga, C. Gartland, C. Petitjean and A. Prochazka, *Purely 1-unrectifiable metric spaces and locally flat Lipschitz functions*, arXiv:2103.09370.